

4.1: Basics

Randomized studies:

- gold standard
- double blind randomized study is the best.

Strengths:

- evaluate causal relationships
- high internal validity
- investigator control over patient exposure.
- prospective data collection
 - standardization of exposure and outcome collection.
- Balance, thru randomization, of known and unknown confounders btwn the groups.

Limitations:

- Higher cost.
- Limited external validity and generalizability
- Ethical considerations
- shorter follow-up
- inefficiency of detection of rare or delayed outcome.

What do we mean by randomization?

- process of assigning of treatment.
- Each participant has the same probability of receiving treatment.

Simple randomization:

Randomization based on a single sequence of random assignments.

In large trials ($n > 200$), that's not a problem.

In small trials ($n < 100$), you might have unequal number of participant in each group.

Block randomization:

Method is used to ensure a balance in sample size across groups over time.

1) TTCC
2) TCTC
3) TCCT
4) CTTC
5) CTCT
6) CCTT

Steps:

- 1) accumulate a block of people (4 people)
- 2) Randomly choose the block assignment order
- 3) Assign the group of block to that order.

Stratified Randomization:

→ Address the need to control and balance covariates.

- 1) Wait for all participants to accumulate.
- 2) Stratify/separate by the covariate(s) (for example, gender).
- 3) Do simple randomization or block randomization for each stratification.

Note: there could be multiple layers of stratification.

Covariate adaptive randomization:

When a new participant is sequentially assigned to a particular treatment group by taking into account the specific covariates and previous assignments of participants.

Cluster (or group) randomization:

The unit of randomization is a group rather than individual.

→ randomize clusters or groups.

What to do when there is non-compliance?

	Took Treatment	Did not take Treatment	
Assigned to treatment group	Compliers (a)	Noncompliers (b)	$a + b$
Assigned to control group	Noncompliers (c)	Compliers (d)	$c + d$
	$a + c$	$b + d$	

What to do when there is non-compliance?

Intent to treat (ITT):

- Analysis compares $(a + b)$ vs $(c + d)$, as assigned,
- Gold standard. Does not break randomization,
- Reduces power.

As Treated (AT):

Analyze data according to the treatment the individual actually received.

- $a + c$ vs. $b + d$.

Per - protocol:

Excludes patients who did not fully comply with treatment protocol.

- compares a vs. d .
- Ignore all non-compliers. But this is problematic as it breaks randomization: non-compliers may differ from compliers in ways that are not measurable.

IV: Instrumental Variable

randomization as IV to adjust for noncompliance.

CACE: complier average causal effect

- Estimate ATE among compliers via principal stratification.
- EM on Bayesian estimation.

4.2: Power and Sample Size:

Power:

- the probability that the analysis will detect an effect when there truly is an effect.
- power = $1 - \beta$, where β = probability of a false negative, or type II error.

Type I error: False positive.

the probability that the analysis will detect an effect when there is no effect.

Denoted by α .

$$\alpha = 0.05.$$

Underpowered, overpowered, just right:

underpowered: your analysis is not able to detect an effect when it is truly there.

- waste of time to do a study.

overpowered: you can detect an effect, but you also detect little things.

- waste of resource.

Effect size = quantitative measure of the magnitude of an effect. The larger the effect size, the stronger the effect. Quantified using Cohen's d .

Cohen's d: used for continuous outcome.

difference btwn 2 means, divided by the standard deviation of the data.

$$d = \frac{\bar{x}_1 - \bar{x}_2}{S}, \text{ where}$$

$\bar{x}_1$ = mean outcome group 1

$\bar{x}_2$ = mean outcome group 2.

$$S = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 + n_2 - 2)}}, \text{ // standard deviation of the data.}$$

n_1 = sample size of group 1

n_2 = " group 2

s_1 = standard deviation group 1

s_2 = " group 2

// this version is the pooled standard deviation.

<u>Effect size:</u>	<u>Cohen's d:</u>
small	.2
medium	.5
large	.8

Design $\Rightarrow$ meaningful effect size

This study has 80% power to detect a small effect size (Cohen's d = 0.2).

$\Rightarrow$ Assuming that there is a true difference of 0.2 or greater in effect size magnitude, the probability that the difference will be detected is 0.8.

Two-sample t-test:

$$H_0: \mu_1 = \mu_2$$

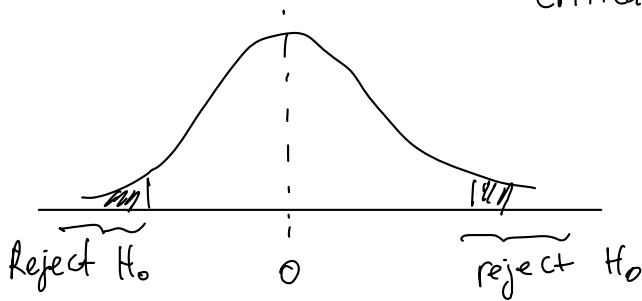
, μ_1, μ_2 = true means

$$H_A: \mu_1 \neq \mu_2$$

$$\Leftrightarrow H_0: \delta = 0, \text{ where } \delta = \mu_1 - \mu_2.$$

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s \sqrt{1/n_1 + 1/n_2}}$$

Reject H_0 if $|t| > \underbrace{T_{1-\alpha/2, v}}_{\text{critical value with } \alpha, v = df}$



$$\text{Power} = 1 - \underbrace{\Phi}_{\substack{\uparrow \\ \text{normal} \\ \text{CDF}}} \left(T_{1-\alpha/2, v} - \frac{\delta^*}{s \sqrt{1/n_1 + 1/n_2}} \right) \quad // \text{ true effect is } \delta^*$$

$\delta = \delta^*$

$$= 1 - \Phi \left(t_{1-\alpha/2, v} - \frac{d}{\sqrt{1/n_1 + 1/n_2}} \right)$$

The moving pieces are power, sample size, and d .

4.3: Power Calculations for a Complex Design / Analysis

Simulation-based Power Calculation:

- Repeatedly simulate your study and calculate the proportion of the studies where the null is rejected.
⇒ this is the estimated power.
- simulate the study under assumption for H_A .
- difficulty: how to simulate your study?
- guess and check
- want the sample size at some fixed power.

Simplify:

Take the complex design and simplify it so that it is like a t-test or chi-square test.

Then use ESS (Effective Sample Size):

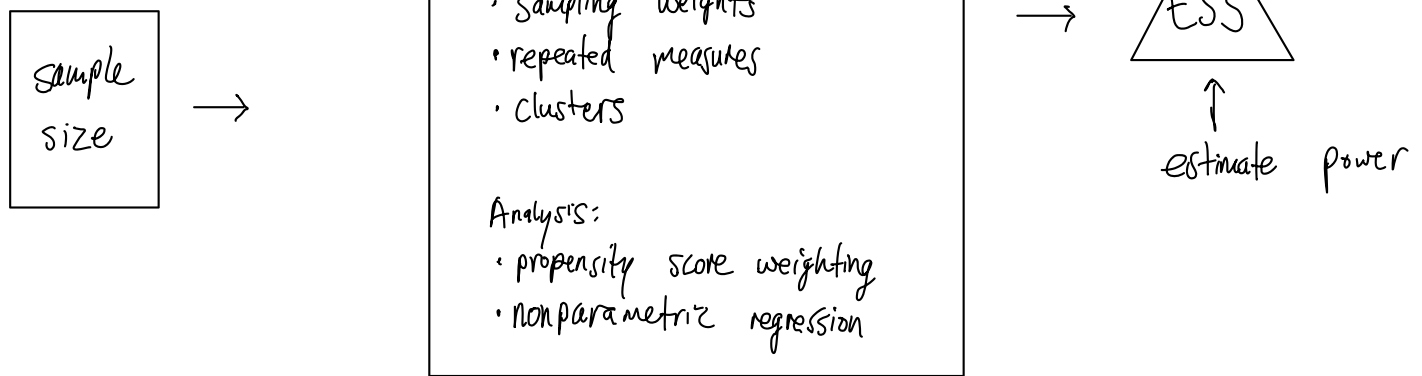
ESS = sample size after you account for complexity in the data structure or in the analysis.

$$ESS = \frac{n}{DEF}$$

↑
design effect

← true sample size

$DEF > 1$, usually. Penalty for sample size.



DEF = quantifies how much the variance has increased because your sample was drawn and adjusted to a specific sampling design, vs simple random sample.

→ quantifies the extent to which your data deviates from a simple random effect.

Many settings and analysis can induce a design effect:

Ex: propensity score weighting, longitudinal data.

Propensity Score

Supposed in your design you are using weight, w_i , for each person i in your sample:

DEF based on Kish's design effect:

$$DEF_w = \frac{\sum_{i=1}^n w_i^2}{\left(\sum_{i=1}^n w_i\right)^2}$$

// Kish's design effect.

Effective sample size:

$$ESS = \frac{n}{DEF_w} = \frac{n}{\frac{\sum_{i=1}^n w_i^2}{\left(\sum_{i=1}^n w_i\right)^2}} = \frac{\left(\sum_{i=1}^n w_i\right)^2}{\sum_{i=1}^n w_i^2}$$

Longitudinal data:

Repeated measure on the same person:

This induces a design effect which is based on the intraclass correlation (ICC).

Recall the ICC $[0, 1]$ describes how strongly measurements in the same group/person resemble each other.

DEF: for such "clustered" data is:

$DEF_c = 1 + (n^* - 1) ICC$, where n^* = size of clusters

$$ESS = \frac{n}{DEF_c} = \frac{n}{1 + (n^* - 1) ICC}$$